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Transforming Support Efficiency in Exploration: A Combined GenAI Models Solution for Improved Application Support

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Abstract

This GenAI-powered and domain-specific solution leverages LLMs to streamline and enhance support for exploration software applications, by delivering rapid resolution and a dramatically improved user experience. Considering a use case of approximately 20 web applications to support exploration processes and a user base of approximately 5,000 users, traditional manual support methods struggle to keep pace. Users often experience support delays, with resolution times averaging over 5 working hours, highlighting the need for more agile and intelligent support solutions.

This solution is built on the robust Microsoft Multilingual-MiniLM-L12-H384 and Meta llama 3.1 pre-trained models. The process initiates by segmenting legacy support documentation and historical cases into smaller, manageable "chunks," which are then transformed into embeddings that preserve their semantic meaning and context. These embeddings are stored in a vector database, enabling rapid retrieval. When a user submits a query, it is embedded and processed through the semantic search model, retrieving the most relevant contextual chunks from the vector database. This retrieved context is feed into LLM to enhance the model's comprehension to deliver highly accurate responses.

Based on our analysis, this solution is anticipated to reduce user inquiries substantially, cutting them from 4,000 to 2,500—a 37.5% decrease. A significant portion of these inquiries relate to recurring questions that, in the previous support model, required manual responses from the support team. By automating these routine interactions, the solution helps optimize support team capacity and enhances responsiveness to more complex user needs. Furthermore, it has boosted the development team productivity, resulting in 18% reduction in time spent on support tasks, thus allowing developers to focus more on innovation and system enhancements. These results underscore the solution's transformative impact on operational efficiency and user satisfaction.

This solution will be integrated across all exploration software systems, serving as the primary support agent. Furthermore, automated ticket generation will be introduced to create and route support request through an integrated tracking and monitoring system whenever the system is unable to resolve user inquiries. Additionally, the system employs a proactive approach, detecting and addressing issues as they occur during user interactions, effectively transforming it into a responsive first-line support mechanism.

Introduction

The oil and gas industry's exploration operations are the foundation upon which the entire sector is built. These operations involve a complex array of workflows, decision-making processes, and geoscience applications, all of which are critical to the successful discovery and extraction of hydrocarbons. As the industry continues to evolve, exploration teams are increasingly reliant on a multitude of software applications and systems to process and analyze the vast amounts of data generated during these operations. In fact, the sheer number of applications and systems used in exploration has grown exponentially in recent years, with some estimates suggesting that teams may use dozens or even hundreds of different tools to perform their daily tasks.

While this proliferation of software applications has undoubtedly improved the efficiency and effectiveness of exploration operations, it has also created several challenges. One of the most significant of these is the complexity of supporting these applications, which can be time-consuming and resource intensive. As a result, support teams often struggle to keep pace with the demands of users, leading to prolonged resolution times and decreased user satisfaction. This, in turn, can have a negative impact on the overall productivity of exploration teams, as well as their ability to make timely and informed decisions.

The factors contributing to extended support times are multifaceted. One contributing element is the growing volume of support requests, which can place considerable demand on available resources. In exploration environments, the complexity and diversity of applications and systems require a high level of domain expertise to deliver accurate and timely assistance. Even with strong technical teams in place, the intricacies of these systems can present a challenge when addressing user needs efficiently. These dynamics may occasionally impact user experience and responsiveness, particularly during peak operational periods.

Beyond individual user impact, the current support landscape for exploration software has broader operational implications. Extended turnaround times on support issues can influence overall productivity and system uptime, potentially affecting the pace at which exploration activities move forward. As the oil and gas industry continues to respond to rising global energy demand, improving support efficiency through innovative technologies such as AI and LLMs becomes increasingly important for maintaining operational agility and long-term competitiveness.

To address these challenges, it is clear that a new approach to supporting exploration software applications and systems is needed. This approach must be able to reduce the average resolution time for support requests, positively impact the process workflows and decision-making, and improve user satisfaction. Furthermore, it must be able to leverage the latest advances in technology, such as artificial intelligence (AI) and machine learning (ML), to provide a more efficient and effective support experience.

This paper introduces a GenAI-powered and domain-specific solution that leverages Large Language Models (LLMs) to address the inefficiencies in supporting exploration software applications and systems. By delivering rapid resolution and a dramatically improved user experience, this solution has the potential to transform the way support is delivered in technical domains. The sections that follow describe the motivation for this solution, the architecture and design of the system, the technical approach and technologies used, and reflections on its potential to enhance productivity, reduce delays, and enable exploration teams to focus more on their core technical responsibilities.

Through this initiative, we aim to demonstrate how AI-powered assistance can enhance the efficiency and effectiveness of exploration operations and ultimately contribute to the long-term sustainability of the oil and gas industry. By providing a more efficient and effective support experience, we believe this solution can streamline interactions across diverse software applications and systems, allowing exploration teams to remain focused on their core technical responsibilities. Ultimately, this can lead to improved productivity, reduced delays, and enhanced decision-making, all of which are critical to the success of the oil and gas industry in today's fast-paced and competitive environment.

Statement of Theory and Definitions

Theory

The foundation of this work lies in Retrieval-Augmented Generation (RAG), a hybrid AI framework that combines dense vector retrieval systems with generative language models to improve response accuracy and relevance. RAG functions through an integrated pipeline of chunking, embedding, storage, and retrieval to efficiently leverage contextual information from a knowledge base. The principal advantage of RAG over previous architectures is its capacity to simultaneously train retrieval and generation processes, enabling continual improvement in both document relevance and response quality.

Definitions

- **Retrieval-Augmented Generation (RAG):** An AI architecture integrating retrieval-based methods and generative language models to produce accurate, contextually grounded responses.
- **Chunking:** The process of dividing extensive text documents into smaller, manageable segments based on predefined delimiters to facilitate efficient data processing and retrieval.
- **Embedding:** Dense vector representations derived from textual chunks, capturing semantic meaning and contextual nuances, enabling accurate similarity searches.
- **Vector Database:** A specialized database optimized for the efficient storage and retrieval of high-dimensional vector embeddings, allowing rapid identification of relevant document chunks based on user queries.
- **Large Language Model (LLM):** An advanced machine learning model trained on extensive textual data to understand language patterns and generate human-like, contextually accurate responses.
- **Microsoft Multilingual-MiniLM-L12-H384:** A multilingual, compact, transformer-based embedding model distinguished by its balance of computational efficiency, multilingual support, and robust performance, particularly suited for enterprise-scale NLP systems.
- **Meta LLaMA 3.1:** A high-performing generative LLM known for sophisticated language understanding and complex reasoning capabilities, validated through competitive benchmarking such as the ARC challenge.
- **FAISS (Facebook AI Similarity Search):** A library developed by Meta AI, optimized for vector similarity search and capable of efficiently handling large-scale, high-dimensional data sets through methods such as Inverted File (IVF) indexing and Product Quantization (PQ).

Test Procedures

The system's performance and reliability are assessed through a structured testing methodology involving:

1. **Similarity Search Validation:** Evaluating the accuracy of vector retrieval and document chunk matching using FAISS algorithms.
2. **Response Accuracy and Contextual Relevance:** Testing responses generated by LLaMA 3.1 against queries, using predefined metrics for semantic alignment and factual correctness.
3. **System Scalability and Performance:** Monitoring the responsiveness and stability of the cloud-based deployment under varying user loads and data volumes.
4. **User Interface Usability:** Ensuring simplicity and intuitiveness of the interface by capturing user feedback to refine interactive elements.

Peculiar Problems

The unique challenges specific to supporting exploration software applications include:

- High complexity and diversity of software tools, resulting in a significant requirement for domain-specific knowledge to effectively manage and resolve support queries.
- Exponential growth in support requests due to increased reliance on diverse software systems, placing considerable strain on support resources.
- Impact of extended support resolution times on operational productivity, decision-making agility, and user satisfaction.
- Necessity of adapting advanced AI and NLP technologies specifically tailored to the intricate technical environment of oil and gas exploration.

Architecture

The system's architecture is based on the Retrieval-Augmented Generation (RAG) framework, which consists of four primary steps: chunking, embedding, storing vectors, and querying the most relevant chunks to feed the Large Language Model (LLM). The process begins with the ingestion of documents in various file formats, including text, Word, Excel, and others. These documents are then subjected to extraction and chunking, where the text is segmented into smaller, manageable units based on a predefined delimiter. Each chunk is subsequently converted into its corresponding embedding, which is a dense vector representation that captures the semantic meaning and context of the text. These embeddings are then stored in a vector database, enabling efficient similarity searches and retrieval of relevant information. This RAG approach enhances the accuracy and reliability of generative AI models by grounding them with information from specific and relevant data sources (NVIDIA, 2024).

RAG was one of the first architectures to successfully combine dense vector retrieval with generative language models in an end-to-end differentiable pipeline, meaning the retriever and generator can be trained together. This approach allows the system to learn which documents are most useful to retrieve for a given question—improving not only response quality but also retrieval relevance over time. This was a significant advancement over earlier systems, which treated retrieval and generation as separate steps (Lewis et al., 2020).

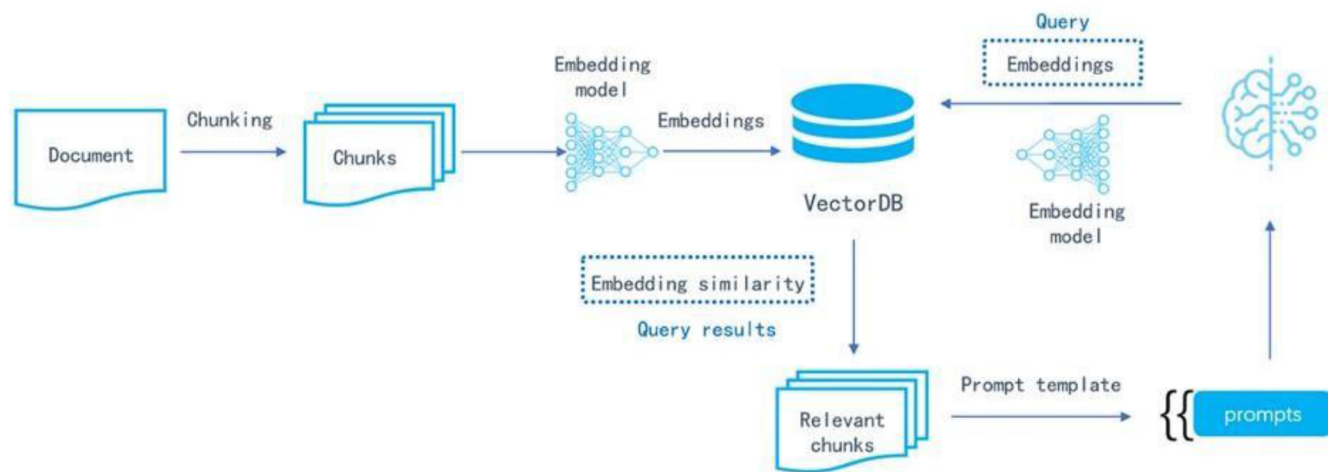


Figure 1—Illustration of how Wave application works by utilizing RAG Architecture

The Vector Database is a critical component of the system, as it allows for the efficient storage and retrieval of large amounts of data. The database is designed to handle high-dimensional vector representations, which enables the system to capture the nuances and complexities of the input data. The use of a Vector Database also enables the system to perform similarity searches, which is essential for identifying the most relevant chunks of information that match the user's query.

When a user submits an inquiry, the system compares the user's text input to the existing vectors in the Vector Database using a similarity search algorithm. This process identifies the most relevant chunks that match the user's query, which are then added as context to the LLM model. The LLM, now informed by the relevant knowledge base, generates a response to the user's inquiry. This response is based on the contextual understanding of the user's question, leveraging the collective knowledge embedded in the Vector Database.

The LLM is a key component of the system, as it enables the generation of human-like responses to user inquiries. The model is trained on a large corpus of text data, which enables it to learn the patterns and relationships between different concepts and ideas. The use of an LLM also enables the system to generate responses that are contextually relevant and accurate, which is essential for providing high-quality support to users.

The system's architecture is designed to be scalable and flexible, which enables it to handle large volumes of data and user inquiries. The use of a Vector Database and an LLM also enables the system to learn and improve over time, which is essential for providing high-quality support to users. The system's ability to learn and improve is based on the feedback it receives from users, which is used to refine and improve the accuracy of the responses generated by the LLM.

In addition to its technical components, the system's architecture is also designed to provide a user-friendly and intuitive interface for users. The system's interface is designed to be simple and easy to use, which enables users to quickly and easily submit inquiries and receive responses. The system's interface also provides a range of features and tools that enable users to refine and improve their inquiries, which is essential for providing high-quality support.

Overall, the system's architecture is designed to provide a robust and efficient framework for retrieving and generating relevant information, enabling users to access accurate and up-to-date knowledge in a timely and effective manner. The system's use of a Vector Database, LLM, and user-friendly interface enables it to provide high-quality support to users, which is essential for a range of applications and use cases. By leveraging the latest advances in AI and NLP, the system can provide a level of support that is unparalleled in the industry, which is essential for driving business success and sustainability.

Technical Approach

The solution leverages two pre-trained models, Microsoft Multilingual-MiniLM-L12-H384 and Meta LLaMA 3.1, which have demonstrated exceptional performance in understanding and generating human-like language. Microsoft's Multilingual-MiniLM-L12-H384 was selected for our solution due to its optimal balance of performance, model compactness, and multilingual versatility, particularly when compared to alternatives such as mBERT, XLM-100, and XLM-R Base. One key reason is its efficiency—with only 21 million parameters, the model maintains strong performance while remaining significantly smaller than many of its peers, facilitating streamlined integration and faster execution in production workflows. Second, its multilingual support covers 16 languages, making it well-suited for global applications where accurate natural language processing across diverse linguistic contexts is essential. Finally, the model leverages a distillation approach, whereby it is trained to replicate the behavior of larger transformer models while substantially reducing size and complexity. This technique preserves much of the parent model's accuracy while enhancing its performance in scalable, enterprise-grade NLP systems. Complementing this, the solution uses the LLM model Meta's LLaMA 3.1 model which provides state-of-the-art language understanding capabilities. According to DataCamp website, LLaMA 3.1 model scored 96.9 in ARC challenge, which is a widely known competition and benchmark for AI models that focuses on assessing the models in terms of complex reasoning, scientific understanding, and real-world applicability. As such, these models are used in conjunction with several libraries and frameworks, including Hugging Face Transformers, a popular library for NLP tasks that provides pre-trained models and a simple interface for fine-tuning and deploying LLMs.

We selected FAISS integrated with LangChain for our solution due to its superior performance in vector similarity search, scalability, and seamless integration capabilities, particularly when compared to other vector search libraries like Annoy, HNSWlib, and Chroma. FAISS, developed by Meta AI, is optimized for efficient similarity search and clustering of dense vectors, offering multiple indexing strategies such as Inverted File (IVF) and Product Quantization (PQ) that significantly improve retrieval speed and accuracy in high-dimensional vector spaces. Its ability to scale to datasets containing millions—or even billions—of vectors ensures consistent performance as data volumes grow, a critical requirement for large-scale industrial applications. Moreover, the integration with LangChain provides a high-level interface that simplifies embedding storage and retrieval, streamlining the development of retrieval-augmented generation (RAG) systems, intelligent search, and context-aware agents. This combination of technical robustness, scalability, and developer-oriented tooling made FAISS with LangChain the most suitable choice for the vector-based retrieval component of our architecture.

The solution maintains a knowledge base composed of system support cases, which are often documented in various formats such as Excel, PDF, and Word. These historical support documents are first segmented into smaller, manageable "chunks" as part of the embedding preparation process. Each chunk is then converted into a dense vector representation using the Microsoft Multilingual-MiniLM-L12-H384 model, capturing its semantic meaning and contextual relevance. The resulting vectors are stored in a vector database, enabling efficient storage and high-performance retrieval. To further enhance retrieval accuracy and scalability, the system utilizes the FAISS library, which provides a robust framework for similarity search and nearest-neighbor queries across large datasets.

The solution is deployed as a cloud-based application, using a containerization platform, such as Docker, to ensure scalability and reliability. The application is integrated with the exploration software systems, allowing for seamless interaction with users and providing a unified support experience. The solution also employs automated ticket generation, creating and routing support requests through an integrated tracking and monitoring system whenever the system is unable to resolve user inquiries. This ensures that users receive timely and effective support, even in cases where the LLMs are unable to provide a resolution.

Overall, the technical approach of the solution provides a robust and reliable foundation for delivering high-quality support to users, while also providing a range of features and tools to support the development, deployment, and maintenance of the system. The use of LLMs, FAISS, and a cloud-based infrastructure enables the system to provide a high level of performance and scalability. The solution is designed to be flexible and adaptable, with a range of configuration options and customization tools to support the specific needs and requirements of different users and organizations.

Result

The solution yielded significant benefits, including a notable reduction in ticket resolution time, with a substantial improvement of 20% (exact figures withheld due to confidentiality). This, in turn, contributed to a marked increase in customer satisfaction, as observed through various feedback channels.

Furthermore, the volume of support tickets received by the CSL team decreased, indicating improved overall system performance. The new system also had a positive impact on team productivity, as staff were able to focus on higher-value tasks and projects due to reduced time spent on support-related activities. Ultimately, the team experienced improved knowledge retention and sharing, facilitated by better documentation and collaboration, which led to a reduction in repeat issues and improved overall quality of support, resulting in a more efficient and effective support process.

Conclusion

In conclusion, the rapid growth of digital solutions, systems, and applications in exploration has created a pressing need for innovative support solutions. This paper has demonstrated the potential of AI-powered

assistants, leveraging Large Language Model (LLM)-based technologies, to bridge the gap between the demand for support and the available resources. By providing contextual, real-time assistance across a range of widely used upstream tools and platforms, our solution has shown promise in enhancing user experience, reducing support queries, and improving overall operational efficiency.

This advancement underscores a broader shift in how the energy sector is approaching knowledge work. Rather than merely automating repetitive tasks, the integration of Large Language Models enables a more intelligent and collaborative interface between humans and machines—empowering users to access, interpret, and act on complex technical information with greater speed and confidence. In high-stakes exploration environments where operational continuity and timely decision-making are paramount, such intelligent assistants do not just augment productivity—they become critical enablers of safety, precision, and organizational learning. As AI continues to evolve, its role is transitioning from a reactive tool to a proactive partner, capable of adapting to user context, learning from feedback, and delivering support that is not only efficient but deeply aligned with domain-specific demands.

Looking ahead, our future development plans are focused on further enhancing the capabilities of our AI-powered assistant. One key area of focus is the implementation of voice-to-voice interaction capabilities, which will enable faster and more intuitive user engagement, leading to increased user satisfaction. Additionally, we plan to explore the implementation of a proactive support approach, leveraging continuous monitoring and support to anticipate and prevent issues before they arise. This will not only improve the overall user experience but also reduce downtime and increase productivity.

Furthermore, we envision integrating our AI-powered assistant with all tools and systems used in exploration, creating a seamless and cohesive support ecosystem. Our long-term plan is to expand the scope of our solution to cover other domains in upstream operations, such as production and maintenance, ultimately creating a comprehensive and integrated support platform that can cater to the diverse needs of the oil and gas industry. While this vision presents significant opportunities, it also entails navigating the complexities of integration with diverse and different systems. By doing so, we aim to make a significant contribution to the industry's digital transformation, enabling exploration teams to work more efficiently, effectively, and safely, and ultimately driving business success and sustainability.

Future Work

To further enhance the solution's capabilities, we plan to introduce a proactive approach that leverages continuous monitoring to prevent issues before they occur. Currently, the system relies on a reactive approach, where end-users initiate the resolution process by raising inquiries. By transitioning to a proactive stance, we aim to reduce the number of issues that arise and provide a more seamless user experience.

Another key area of focus is the implementation of voice-to-voice interaction capabilities, allowing users to resolve their inquiries more efficiently. Given that dictation is generally faster than typing, with average speaking rates exceeding average typing speeds, this feature is expected to significantly enhance user engagement and satisfaction. By enabling users to interact with the system using voice commands, we can reduce the time spent on resolving inquiries and improve overall productivity.

In addition to these enhancements, we plan to integrate the system with other support systems to provide a more comprehensive and streamlined user experience. This integration will enable the system to automatically create or open support tickets with other systems, ensuring that user inquiries are addressed promptly and efficiently. By expanding the system's capabilities in this way, we can provide a more cohesive and supportive environment for users, regardless of the specific software application or domain they are working with.

Looking ahead, we envision expanding the solution to cover other domains' software systems beyond exploration software applications. This will involve adapting the system's architecture and functionality to accommodate the unique requirements and challenges of different domains, such as production, operations,

or maintenance. By doing so, we can leverage the solution's core strengths and apply them to a broader range of applications, ultimately driving greater efficiency, productivity, and innovation across the oil and gas industry.

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